MEMORANDUM · MAIN DOCUMENT 3 OF THE SITUATIONAL-PICTURE SERIES

AI and the Situational Picture – From Report to Situation Service

How Artificial Intelligence Changes the Creation and Provision of Situational Information

Version 1.2 · As of 2026-08-14 · Audience: decision-makers in an enterprise organization – no AI, military or agile background required

About this document: Curated, conceived and editorially owned by **Robert Seebauer**; researched and produced with the support of **Claude** (Anthropic – model **Claude Fable 5**, via Claude Code). This is **version 1.2** (reference maintenance for the online publication – cross-references and freely available sources now carry real links, working-archive paths removed; content unchanged; the deepened v2.0 after evaluation of *Prediction Machines*, *Co-Intelligence* and *Noise* remains planned). Sister documents: "Team of Teams & Staff Training" (people & rituals) and "Enterprise Architecture & Solution Situational Picture" (content & structure) – together with this memorandum (tooling & provision) the triad of the series. Earlier versions: release archive `denkschriften/<date>`.

Executive Summary

Starting question: The series has shown *how* a situational picture is run (rituals, staff training) and *what* belongs in it (capabilities, debt, confidence). What remains open is the question of tooling: **To what extent can Artificial Intelligence help with creating a situational picture and – above all – with the provision of situational information?**

THE SINGLE MOST IMPORTANT SENTENCE IN THIS DOCUMENT

AI makes situational information cheap while judgment remains scarce – therefore AI belongs on the staff, not on the commander's hill, and its greatest lever is **provision**: the right information, at the right time, in the language of the recipient.

Findings in five sentences:

1. **The economics reverse.** AI drastically lowers the cost of information processing and prediction; in return, the value of the complementary good rises – human judgment (Agrawal/Gans/Goldfarb). The bottleneck of the situational picture migrates from *creation* to *interpretation and delivery*.
2. **The military supplies the mature model.** For decades, the intelligence cycle has treated provision ("dissemination") as *a phase of its own with a doctrine of its own* – and the AI practice of NATO and armed forces shows a clear line: AI condenses situation and analysis, decision and responsibility stay with the human.
3. **The greatest civilian lever is the situation service.** For the first time, AI makes a *standing, queryable* situation service for the portfolio affordable – the shift from the periodic report to a service you can query at any time and that reports in on its own at decision points. The Security Operations Center has been proving for years that companies can run such a thing.
4. **No provenance, no truth.** An AI-generated situational picture without a statement of sources and confidence is the perfect "Information Masquerade" (Milchman). The empirical record confirms both – ~90% AI adoption, but ~30% distrust of AI output, and AI *amplifies* what is there instead of repairing it (DORA 2025). Six principles draw the guardrails from this.

5 **Consequence for us:** A new competency field 7 (“AI-assisted staff work”) and curriculum module 7, a 90-day pilot with the delta briefing as the first step – and the software roadmap is extended by **Phase D (AI assistance)** behind the deterministic core.

What “AI” means here (taxonomy, applies to the whole document)

Level	What	In our context today
1 · Rule-based automation	Deterministic pipelines, thresholds, traffic lights	The entire report pipeline
2 · Statistics & classical ML	Monte Carlo forecasts, anomaly detection, fault prediction	Simulate module; *Fault Detection*
3 · Generative AI / LLMs (focus)	Summarizing, translating (languages **and** readings), Q&A, drafts	Memoranda production, manuals in 5 languages
4 · Agentic AI	Autonomous collection, multi-step assignments, tool use	Outlook only (D.2, guardrails)

Anyone deciding on “AI in the situational picture” should always be able to say which level they are talking about – opportunities, risks and guardrails differ considerably by level.

Part A – The sources: AI in our own corpus

The four AI titles of the 22-volume Apress corpus (full texts in the curator's working archive), read through the situational-picture lens – plus Milchman as the hinge to the EA memorandum.

A.1 *Architecting Enterprise AI Applications* – drawing the strategic boundary

The strategic-conceptual work of the corpus on the question of **where AI should decide and where it should not**: human-vs.-AI division of labor, meta-systems, overfitting and proxy traps. Core stance: deploy AI deliberately *within limits* where context and responsibility matter.

For us: the corpus' principal witness for the role model of this memorandum – AI as a staff member with a defined mission, not as a decision-maker.

A.2 *Mastering Machine Learning Architecture and Solutions* – the lifecycle

ML systems as architecture building blocks with a **lifecycle** of their own: data pipelines, model selection, operations, drift – MLOps as “DevOps for models”.

For us: Whoever builds AI into the situational picture takes on operational responsibility (risk catalogue D.2: model drift); AI building blocks are not a one-time installation.

A.3 *AI and Microservices* – the applied view

AI-supported development and API/testing work in microservice landscapes.

For us: Evidence that level-3 AI already has tool status in everyday engineering – the question is not *whether* but *with which guardrails* (congruent with the DORA findings in A2.5).

A.4 *Fault Detection in Microservice Architectures* – analysis in pure form

Data-driven **fault prediction** from structural metrics (coupling/cohesion → fault probability, evaluated with precision/recall).

For us: the concrete pattern for turning existing metrics into *leading indicators* – level-2 AI for the analyze phase of the cycle (C.1), long before an LLM enters the picture.

A.5 The hinge: Milchman, *Unit Oriented Enterprise Architecture*

The only work in the corpus that literally speaks of **“situational awareness”** – subtitle: *Constructing Large Sociotechnical Systems in the Age of AI*. In the AI age, his diagnosis of the **“Information Masquerade”** (information flows worse than everyone claims) becomes a double warning: AI can *fight* the masquerade (provenance, confidence, availability) or *perfect* it (fluent text without substance). Which side wins is decided by the principles in C.4.

Part A2 – Supplementary evidence: economics, automation research, empirical findings

*Six sources outside the corpus, each carrying one flank. The books marked with * have been acquired but not yet evaluated – their core theses are sketched from general knowledge and will be deepened in v2.0.*

A2.1 Agrawal/Gans/Goldfarb – *Prediction Machines** (2018/2022): the economics

The thesis of the Rotman economists: economically speaking, AI is a **price collapse for prediction** (in the broad sense: filling in missing information). When a good becomes cheap, the value of its **complements** rises – for prediction these are data and above all **judgment** (assessing what a prediction *means* and what is to be done).

For us: the economic justification of the single most important sentence. The situational picture becomes cheaper to produce; what becomes more valuable is supplying the scarce judgment of decision-makers properly – in other words, provision.

A2.2 Mollick – *Co-Intelligence** (2024): the practice patterns

The Wharton professor distills everyday work with LLMs into simple rules – among them *"be the human in the loop"* and the warning about the *"jagged frontier"*: the capability boundary of LLMs runs jagged; they are surprisingly strong at tasks that look hard, and surprisingly weak at tasks that look easy.

For us: This is exactly why AI staff work needs *training* (C.5) – the jags of the frontier can only be learned through guided practice, not from slides.

A2.3 Bainbridge – "Ironies of Automation" (1983): the rebuttal

The classic of automation research (doi.org), in four pages: automation takes over the *easy* parts of a task and leaves the human the *hardest* ones – monitoring and exception handling – while at the same time depriving them of the daily practice they would need for exactly that. The better the automation, the **more important and at the same time worse prepared** the human.

For us: perhaps the most important single warning in this memorandum – principle 5 "The human keeps practicing" (C.4) and the AI-free katas in the curriculum (D.4) are the direct answer.

A2.4 Parasuraman/Sheridan/Wickens (2000): the dial

The standard model of human-automation research distinguishes **four functional classes** – information acquisition, information analysis, decision selection, action implementation – and, per class, **levels of automation** from "the human does everything" to "the machine decides autonomously". The point: the level of automation is **to be chosen separately per functional class**.

For us: the scientific foundation of the automation dial in C.2 – "staff member, not commander" becomes a tunable setting instead of a ban.

A2.5 DORA 2025 – *State of AI-assisted Software Development*: the empirical findings

The annual DORA study (Google Cloud, ~5,000 respondents worldwide) delivers three findings for decision-makers: (1) **~90% of software professionals use AI** at work – the adoption question is settled. (2) The **"amplifier effect"**: AI amplifies existing strengths and weaknesses; high-performing organizations get better, weak ones see their problems more clearly – AI *repairs* nothing. (3) Despite broad usage, **~30% report little or no trust** in AI-generated output; the greatest returns come not from the tools but from platform quality and clear workflows.

For us: (2) is the empirical confirmation of the ToT single sentence ("ritual before tool") transferred to AI; (3) makes provenance and confidence a question of acceptance, not of style.

A2.6 Duke – *Thinking in Bets* (2018): the hygiene of judgment (new in v1.1)

Unlike the titles marked with *, already evaluated. The former poker professional addresses the act of evaluating under uncertainty itself – three contributions carry directly: **resulting** (separating decision quality from outcome quality) is the literacy without which probabilistic situational information does not survive politically – a 15% scenario that materializes does not refute a P85 forecast. **Truthseeking groups** (a truthseeking charter: sharing information, organized skepticism, accountability) supply the *social* antidote to automation bias – the group cross-check before an AI suggestion is adopted. **Ulysses contracts** (advance commitments that bind the later self) give the decision-point wake-up call (M4) its foundation.

For us: Duke trains exactly the judgment that, per *Prediction Machines*, is the scarce good. (Full appraisal: ToT memorandum, Part A2.5.)

*Additionally planned for v2.0: Kahneman/Sibony/Sunstein, **Noise*** – judgment scatter as an error source of its own alongside bias; relevant to the question of where AI support can make human judgments more consistent without replacing them.*

Part B – Reference practice: military intelligence and its AI

The method of the series: first understand the mature outside discipline, then transfer. For situational information, the most mature system is military intelligence – it has been answering our guiding question institutionally for decades.

B.1 The intelligence cycle in plain language

US doctrine (Joint Publication 2-0, *Joint Intelligence*) organizes intelligence work as a cycle of **planning/direction → collection → processing → analysis → dissemination/integration**, with constant evaluation and feedback. Three properties make it valuable for us:

1. **It starts with the decision-maker, not with the data**
   At the beginning stands the leader's prioritized information need (**CCIR/PIR** – Commander's Critical / Priority Intelligence Requirements): *Which information would change my decision?* That is exactly Hubbard's EVI question in doctrinal form – what gets collected is what can change decisions, not what is measurable.

2. **Dissemination is a phase of its own with a doctrine of its own**
   The best analysis is worthless if it does not reach the decision-maker in time, comprehensibly and in the right format – which is why the doctrine explicitly regulates *push* (pre-planned delivery to defined recipients) and *pull* (access to holdings on demand), recipient fit and timeliness. **Exactly this phase is what our guiding question addresses with “above all the provision”.**

3. **It is a cycle**
   Every decision creates new information needs; evaluation runs continuously alongside. A situational picture is never “finished”.

B.2 What the doctrine teaches about provision

From dissemination doctrine and staff practice (cf. the Team-of-Teams memorandum, Part B), five provision lessons can be distilled that hold independently of AI – AI merely makes them affordable:

- **BLUF – Bottom Line Up Front:** core message first, derivation afterwards; the situation briefing is a practiced form, not a piece of prose.
- **Recipient-appropriate:** the same facts are prepared differently per level (commander ↔ specialist staff) – *without* the facts changing.
- **Push by plan, pull on demand:** critical items are actively delivered (at decision points), everything else is available on call.
- **Timeliness before perfection:** a timely 80% picture beats a perfect one from yesterday (McChrystal's O&I lived on this principle).
- **Provenance and reliability are part of the report:** intelligence has always rated its sources explicitly – the civilian counterpart is our confidence flags (EA principle 4).

B.3 AI in the alliance: what NATO and research report – and what stays with the human

- **NATO ACT, *Decision-Making in the Age of Big Data and AI*** (2021): Multimodal sensor data plus analytics are to enable a **consolidated, multidimensional situational picture in near real time**; AI relieves analysts of sifting and fusion work and accelerates decision cycles. Just as clearly, the report names the preconditions: data quality, interoperability, user trust – and human responsibility.
- **RAND (RRA4316-1)** analyzes AI along four fundamental military competitions – mass vs. quality, **hiding vs. finding**, **centralized vs. decentralized command**, cyber offense vs. defense. Two are central for us: AI is primarily a *finding technology* (it lifts signals out of noise – that is situational-picture work), and it can strengthen **both** command

models. That is a warning: whoever *can* see everything centrally with AI will be tempted to *decide* everything centrally again – McChrystal's coupling (situational picture **plus** decentralized authority) must be actively defended against this recentralization reflex.

- **Atlantic Council** (report on NATO's AI integration): AI as an enabler of consolidated multi-domain situational pictures in C4ISR – the same thrust from the alliance perspective.
- **NATO itself draws the line:** its *Principles of Responsible Use* (2021) demand, among other things, accountability (humans remain answerable), traceability and governability of AI deployments. Translated into our image: **AI serves on the staff – command it does not.**

B.4 The civilian proof of existence: the Security Operations Center

Companies have long been running a standing situation service – where the fire is: the **SOC** maintains a cyber situational picture around the clock. Sensors deliver events, correlation systems (SIEM) condense them, **AI-supported triage** prioritizes, alerts arrive with context, and a human analyst decides on the response. Two lessons from twenty years of SOC practice transfer directly: **alert fatigue** is the death of any situation service (hence prioritization by decision relevance – EVI), and **automation ends before the response** – even in the SOC, the machine rarely triggers action on its own. For the portfolio, where millions are decided, nothing comparable exists – not for lack of need, but because a humanly staffed situation service would be unaffordable. **AI changes that math.**

Part C – Transfer: the AI-supported situation service for the enterprise

C.1 The cycle for the enterprise situational picture – and the four roles of AI

The intelligence cycle maps directly onto the situational-picture work of the solution/portfolio level (and onto our software pipeline):

DIRECT What must the decision-maker know? (PIR/CCIR, civilian: define information needs by EVI)

↓

COLLECT Jira, Git, CI/CD, monitoring, registers `[get_data]`

↓

PROCESS pool, canonicalize, flag quality `[transform]`

↓

ANALYZE trends, anomalies, forecasts, scenarios `[build_reports, simulate]`

↓

PROVIDE readings, delta, Q&A, alerts, situation briefing ← **focus** `[report + situation service]`

↓

DECIDE & FEED BACK sync/O&I, AAR, new questions → back to DIRECT `[ritual + human]`

Across the cycle, AI's contribution can be captured in **four roles** – all four are staff roles, not command roles:

Role	Cycle steps	Examples	Maturity
1 · Sensor & collector	Collect, process	Extraction, classification ("which capability does this epic affect?"), duplicates, justifying quality flags	productive today
2 · Staff aide (editor)	Process, provide	Summaries, translation between readings, delta briefings, situation-briefing drafts, multilingual output	productive today, with mandatory review
3 · Analyst & sparring partner	Analyze	Anomaly/trend detection, forecasts (MC/ML), scenario drafts for wargaming, premortem questions, attacking assumptions (red team)	partly productive, partly worth piloting
4 · Situation service	Provide	Queryable situational picture, decision-point wake-up calls, audience-appropriate delivery	worth piloting – the new territory

C.2 The automation dial: how much AI per step?

Following Parasuraman/Sheridan/Wickens (A2.4), the level of automation is set **per functional class** – for the situational picture we recommend four dial settings:

1. **Information acquisition and processing: dial high**
   This is where the cost reduction happens; human drudgery in data collection is not a mark of quality but waste.

2. **Analysis: dial medium to high**
   With a confidence statement on every result and spot-check verification as staff routine.

3 **Decision selection: dial low**

AI may draft, compare and recommend options – the choosing is done by humans (which is also the MDMP logic: the staff develops options, the commander decides).

4 **Execution: only with human release**

In the situational-picture context this means in practice: no automatic escalations or reprioritizations without an agreed rule and a named responsible person.

This turns "AI is a staff member, not a commander" from a metaphor into a tunable setting – differentiated enough for real architecture decisions, simple enough for the board slide.

C.3 The situation service: seven provision patterns (the centerpiece)

The emphasis of the guiding question lies on provision – here are the seven patterns with which AI turns the report into a situation service:

#	Pattern	What it delivers	Main risk	Guardrail
M1	**Audience-specific readings**	Executive BLUF and engineering detail from the **same** numbers; extends EA principle 3 to "one picture, n readings"	Personalization fragments the single truth	One data basis, a shared core of metrics, provenance per statement
M2	**Delta briefing**	"What has changed since the last sync – and what is accelerating?" (Hohpe's "first derivative" as a product)	Noise gets sold as change	The deterministic diff supplies the facts, AI only the wording
M3	**Queryable situational picture**	Q&A in everyday language over the report data; drill-down without a tooling hurdle – more people can *read* the situational picture	Hallucination: eloquent answers without a basis	**Citation requirement** – no answer without a source reference to the data path; "I don't know" is permitted
M4	**Decision-point wake-up call**	Leading indicator + threshold + agreed response; AI monitors and reports **with context** instead of bare alerts	Alert fatigue (SOC lesson)	Thresholds and responses are agreed in the ritual, not invented by the AI – a **Ulysses contract** (Duke): The board binds in advance its later self in the heat of the moment
M5	**Audience-appropriate delivery**	Role, channel, timing, **language**; including the **situational-picture subscription**: periodic role-specific digests as a plannable push form	Delivery illusion: "delivered" ≠ "understood"	Backbrief spot checks in the ritual
M6	**Situation-briefing draft**	AI drafts the BLUF briefing for the sync; the human reviews, shortens, presents	The ritual degenerates into a reading-out of AI text	The presenter signs responsible; draft ≠ briefing
M7	**Institutional memory**	AAR archive, decision/assumption log searchable and queryable – "Have we seen this before? What did we decide back then – and why?"	Outdated decisions as current truth	Every answer carries the date and validity status of its source

> **Round-the-clock availability vs. ritual.** Does the 24/7 situation service devalue the portfolio sync? No – it upgrades it: if the information reached everyone beforehand, the sync turns from a reporting meeting into a **decision meeting** (McChrystal's O&I was, precisely, *daily*). The only danger is cancelling the ritual "because it's all in the chat anyway" – then the stage dies on which leadership demonstrates culture (ToT success factor 2: rituals before tools).

> **One picture, n readings.** How much personalization can the single truth bear? Readings may differ in selection, depth and language – **never in the numbers and never in the risk picture**. The shared core (traffic lights, top risks, confidence) cannot be personalized; whoever personalizes it builds watermelon reports with AI acceleration.

The third tension – radical transparency vs. protection of employees – is so fundamental that it is answered not as a trade-off but as a principle: see P6.

C.4 The six AI situational-picture principles

Designed to connect to the five situational-picture principles of the EA memorandum – and deliberately extended by a sixth:

1. **AI is a staff member, not a commander.**
It collects, drafts, translates and contradicts – it does not decide and is accountable for nothing. Responsibility cannot be delegated to something that can bear no accountability. (ToT B.1; NATO principles; Prediction Machines.)

2. **Provenance before eloquence.**
Every AI statement in the situational picture carries source and confidence – otherwise it is the perfect Information Masquerade. (Milchman; EA principle 4; DORA trust gap.)

3. **The LLM writes, it does not calculate.**
Numbers originate in the deterministic core (pipeline, simulation); the AI phrases, translates and explains them. A freely invented number is a system defect, not a cosmetic one. (Our own architecture; Hubbard calibration.)

4. **To provide is to prioritize.**
What gets delivered is what can change decisions (EVI) – otherwise AI merely automates the KPI graveyard and produces alert fatigue. (Hubbard; SOC lesson; Goodhart warning.)

5. **The human keeps practicing.**
Whoever leaves every draft to the AI loses the competence to spot its errors – training, katas and AI-free exercises are the protection against deskilling. (Bainbridge; Marquet "competence"; Martin "practice".)

6. **The situational picture knows teams, not individuals.**
Aggregation at team/ART level is a hard rule, not a stylistic device: no drill-down to individuals, no individual measurement – all the more so once AI makes such analysis cheap. At the same time the most effective compliance simplification (co-determination, data protection, AI regulation). (Standing line of the series; legal review box D.2.)

C.5 Competency field 7: AI-assisted staff work

The Team-of-Teams memorandum defines six competency fields for portfolio staffs. AI adds a seventh – **not** "prompt tricks" but staff craft:

- **Being able to task:** formulating precise assignments for AI (context, sources, format, limits) – the civilian form of drafting an order.
- **Being able to verify:** running spot checks, checking source references, holding numbers against the deterministic core; knowing *when* full review is required (decision papers) and when a spot check suffices (routine texts).

- **Reading and writing confidence:** understanding calibrated uncertainty language and demanding it from AI (connecting to the calibration training from curriculum module 3); evaluating forecasts without “resulting” (Duke): A 15% scenario that materializes does not refute a P85 forecast.
- **Recognizing automation bias:** knowing one's own tendency to follow fluent suggestions under time pressure – and mastering counter-rituals (four-eyes principle, anonymous advance votes *before* the AI suggestion, the truthseeking cross-check in the group per Duke).
- **Mapping the jagged frontier:** learning through guided practice where AI is strong in one's own context and where it is unreliable (Mollick) – this knowledge goes stale with every model change and must be maintained.

Part D – Implementation

D.1 Fields of application by maturity (decision-maker matrix)

Field of application (role)	Maturity	Benefit	Main guardrail
Data extraction & classification (role 1)	**productive**	Collection effort drops drastically	Quality flags instead of blind trust
Summaries, readings, translation (role 2)	**productive with mandatory review**	Rosetta-Stone work becomes affordable	Principles 2 + 3; mandatory review before dispatch
Anomaly/trend/forecast (role 3, level 2)	**productive** (statistics)	Leading indicators instead of the rear-view mirror	Confidence intervals instead of point values
Scenario/premortem assistance (role 3, level 3)	**worth piloting**	Red-team capacity for every staff	Human moderates; AI supplies raw material
Queryable situational picture, wake-up calls, delivery (role 4)	**worth piloting**	From report to situation service	Citation requirement; EVI prioritization; P6
Agentic collection/action (level 4)	**outlook**	In prospect: standing research assignments	Only with execution approvals (C.2, dial 4)

D.2 Risks & governance

Risk	Mechanism	Countermeasure
Hallucination	The LLM invents plausible details where data is missing	Citation requirement (M3), generation only from supplied data, rewarding "I don't know", principle 3
Automation bias	AI suggestions are adopted uncritically under time pressure	Confidence statements, spot-check routine, four-eyes review for decision papers, truthseeking cross-check (Duke)
Deskilling	Automation withdraws the very practice that exceptions require (Bainbridge)	AI-free katas (D.4), a "do the math yourself once" rotation, principle 5
False certainty	Perfectly worded text suggests certainty in the Clausewitzian fog	Confidence belongs to the number *and* to the wording; standardized uncertainty language
Goodhart amplification	Teams optimize for AI-readable signals once AI reads and writes reports	A metric set instead of a single metric; qualitative cross-checks in the ritual
Confidentiality & data leakage	The situational picture bundles sensitive company information; external AI services as an outflow channel	Explicit deployment decision (on-prem / EU cloud / zero retention); data classification **before** connecting AI
Prompt injection	Manipulated content in source data steers the answers of the queryable situational picture	Treat inputs as data, separation of privileges, answer provenance
Model drift & operations	AI building blocks have a lifecycle of their own (MLOps)	Name operational responsibility; monitoring of the AI components themselves

WARNING BOX – THE MASQUERADE TRAP

An AI-generated situational picture without proof of provenance is the perfect Information Masquerade: fluent, convincing, unverifiable. It feels like an information gain and is a loss of control. Whoever adopts only a single rule from this document, let it be the **citation requirement**.

Legal review box (have this legally reviewed at introduction; information as of mid-2026): EU AI Act – staggered entry into application: prohibitions (including emotion recognition in the workplace) since early 2025, obligations for general-purpose AI models since mid-2025, high-risk obligations phase in from 2026/2027. Critical: systems relating to the performance/conduct of employees may need to be classified as **high-risk** (Annex III, employment context). On top of that, nationally: **§ 87 Abs. 1 Nr. 6 BetrVG** (German Works Constitution Act: co-determination for technical monitoring of performance/conduct) and the **GDPR**. Principle 6 ("teams, not individuals") is the most effective simplification of all these questions – whoever evaluates no personal data has half the review behind them. Involve the works council early, not after the fact.

D.3 The 90-day pilot

1. **Weeks 1–2 – Direct:** survey the decision-makers' information needs by EVI ("Which information would change which decision?"); make the data-classification and deployment decision for AI services; inform the works council (P6 as a commitment).
2. **Weeks 3–6 – first pattern productive:** introduce the **delta briefing (M2)** – deterministic report diff plus AI wording, sent to all participants before each sync. Lowest risk, immediately tangible, a clean demonstration of principle 3.
3. **Weeks 7–12 – new territory, controlled:** the **queryable situational picture (M3)** as a supervised pilot for a defined user group, with citation requirement and logging; in parallel, competency-field-7 training for the staff.
4. **Wrap-up – a retro on the situation service itself:** Do users trust the answers – and *why* (not)? Which questions could the situational picture not answer (= gaps in the data)? Decision on expansion (M4, M5, M7).

D.4 Curriculum module 7: AI-assisted staff work

Extension of the 12-month curriculum of the Team-of-Teams memorandum (there Part D.2) by a seventh module – alongside the job, exercise-heavy:

- **Content:** the four roles and the automation dial; tasking/verifying/confidence (C.5); risk catalogue and legal situation (D.2); the jagged frontier in one's own context.
- **Exercises:** (1) produce the same situation briefing once with and once without AI, and compare; (2) audit a prepared AI briefing with built-in errors ("find the hallucination"); (3) moderate a premortem with AI raw material; (4) a monthly **AI-free kata** – a situation analysis in 30 minutes by hand, so the form sits when the tool fails (the Bainbridge answer).
- **Artifact:** a one-page **AI usage policy for staff work** (which level for what, review duties, citation requirement, P6) – the governance document most organizations do not yet have.

D.5 The purchasable market (vendor-neutral, by category)

- **Enterprise LLM platforms** with corporate data protection (private instances, EU options, zero-retention contracts) – the basis for roles 2 and 4.
- **BI/analytics copilots** (natural-language querying of existing dashboards) – the fastest route to M3-like functionality, but check the citation requirement and the data path.
- **AIOps/anomaly tools** (operational data, level 2) – mature; deliver leading indicators for the technical health picture (connecting to the SLO/DORA dimension of the EA roadmap).
- **Forecasting tools** (Monte Carlo/probabilistic) – already present as our own Simulate module; external ones only needed if third-party data sources are added.

- **AI competence trainings** – buy them tailored to staff work (verification, bias, law), not as a prompting course; for method training (red teaming, calibration, superforecasting) see the program table of the Team-of-Teams memorandum D.1.

Connectivity to our software (memo items)

The roadmap of the `portfolio` module (`docs/portfolio_roadmap_solution-lagebild.md`) is extended by **Phase D – AI assistance**, strictly **behind** the deterministic core:

#	Feature	Pattern	Prerequisite
D1	LLM executive summary (numbers from `summary.py`, LLM does the wording)	M1/M6	–
D2	**Delta briefing** (deterministic report diff + narration) ⟵ *first pilot*	M2	two report snapshots
D3	Queryable situational picture (Q&A over report data with citation requirement)	M3	provenance model
D4	Anomaly notices (statistics in the core, AI explains)	M4	–
D5	Red-team assistant (premortem questions from the assumption/decision log)	Role 3	Roadmap B4
D6	Multilingual report output	M5	–

Architecture principle: The AI layer is optional and replaceable (own model/own key, local or API); the core remains able to run offline and true to the stdlib. **No core feature may require AI – AI refines, it does not carry.**

> **Reflexive evidence:** This series practices what it describes. Every memorandum is produced with AI support and carries the provenance note (versioning ground rule 5 of the series); the software's manuals appear AI-translated in five languages; the forecasts are computed by a deterministic Monte Carlo module whose results AI may explain but never generate. We do not write about AI staff work – we run it, with exactly the guardrails this document demands.

Connectivity to the series

- **ToT memorandum:** The O&I ritual remains the decision forum – the situation service supplies it; empowered execution is defended against the recentralization reflex (B.3/RAND); competency field 7 and module 7 dock onto the curriculum and the competence catalogue.
- **EA memorandum:** The six AI principles continue the five situational-picture principles; the EA dimensions (capability, NFR, ROAM, dependencies) are the data basis the situation service makes queryable; with M2/M4 the control loop gets its pacemakers.
- **Architecture Thought Leaders companion memorandum:** Hohpe's "first derivative" becomes a product as the delta briefing; Fowler's pattern vocabulary underpins the shared language of the AI usage policy; Martin's culture of practice carries principle 5; Hohmann's rituals remain the human stage on which AI raw material is negotiated.
- **Situational-awareness reading list** (in the curator's working archive): takes up the new sources on AI & situation service and on judgment (Duke).

Bibliography

Own corpus (full texts in the curator's working archive): *Architecting Enterprise AI Applications* · *Mastering Machine Learning Architecture and Solutions* · *AI and Microservices* · *Fault Detection in Microservice Architectures* (all Apress) · Milchman: *Unit Oriented Enterprise Architecture – Constructing Large Sociotechnical Systems in the Age of AI*. Apress, 2024.

Newly archived (freely available, PDFs in the curator's working archive):

- NATO ACT: *NATO Decision-Making in the Age of Big Data and Artificial Intelligence* (2021) – act.nato.int
- RAND: *How Artificial Intelligence Could Reshape Four Essential Competitions in Future Warfare* (RRA4316-1) – rand.org
- Bainbridge, Lisanne: "Ironies of Automation". *Automatica* 19(6), 1983 – doi.org.

Online (cited):

- DORA: *State of AI-assisted Software Development* (2025) – dora.dev/dora-report-2025 (download via form)
- US Joint Chiefs of Staff: *Joint Publication 2-0, Joint Intelligence* – jcs.mil/Doctrine (public edition)
- Atlantic Council: *How NATO can integrate AI to prevail in future algorithmic warfare* – atlanticcouncil.org
- NATO: *Summary of the NATO Artificial Intelligence Strategy / Principles of Responsible Use* (2021) – nato.int
- Parasuraman/Sheridan/Wickens: "A Model for Types and Levels of Human Interaction with Automation". *IEEE Trans. SMC–A* 30(3), 2000 – doi.org.

Additionally evaluated (new in v1.1): Duke, Annie: *Thinking in Bets. Making Smarter Decisions When You Don't Have All the Facts*. Portfolio, 2018 — Part A2.6; full appraisal: ToT memorandum (A2.5).

Being acquired (deepening in v2.0): Agrawal/Gans/Goldfarb: *Prediction Machines* (updated ed. 2022) · Mollick: *Co-Intelligence* (2024) · Kahneman/Sibony/Sunstein: *Noise* (2021).

Internal references: "Team of Teams & Staff Training" · "Enterprise Architecture & Solution Situational Picture" · "Architecture Thought Leaders" · decision record in the curator's working archive (decision state 2026-08-13) · situational-awareness reading list (in the curator's working archive) · `docs/portfolio_roadmap_solution-lagebild.md` (Phase D).

Version history

Version	Date	Changes
1.0	2026-08-13	Initial edition per the decision state of the discussion paper of 2026-08-13: dual guiding image (staff member/situation service), four-level taxonomy, Part B military intelligence + NATO/RAND + SOC, seven provision patterns, six AI principles (incl. P6 "teams, not individuals"), automation dial, competency field 7, 90-day pilot, curriculum module 7, software Phase D (pilot: delta briefing). Planned v2.0: deepening after evaluation of *Prediction Machines*, *Co-Intelligence*, *Noise*.
1.1	2026-08-13	Minor addition (no reorientation): Annie Duke, *Thinking in Bets*, worked in as A2.6 – resulting literacy for probabilistic situational information (C.5), the truthseeking cross-check against automation bias (C.5, D.2), the decision-point wake-up call (M4) classified as a Ulysses contract (C.3); sister-document reference updated to ToT v4.0. The deepened v2.0 remains planned.
1.2	2026-08-14	Reference maintenance for the online publication: cross-references to the series link the online overview, freely available sources carry real URLs (DOI/official pages), working-archive paths removed; cross-references without version numbers. Content unchanged.

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